

- LO1:** Identify the characteristics and components of academic research
- LO2:** Analyse research problems in various IT fields
- LO3:** Evaluate research activities for ethical/unethical conduct
- LO4:** Identifying different sources of information
- LO5:** Evaluate the quality/reliability of different sources
- LO6:** Apply referencing styles in the acknowledgement of sources
- LO7:** Evaluate different research strategies
- LO8:** Evaluate different sampling methods for specific research scenarios
- LO9:** Evaluate different data collection methods for specific research scenarios
- LO10:** Evaluate different data analysis methods for specific research scenarios

QUESTION 1: MULTIPLE-CHOICE [30]

1. Answer: LO1: C [2]
2. Answer: LO2: D [2]
3. Answer: LO3: A [2]
4. Answer: LO4: B [2]
5. Answer: LO5: C [2]
6. Answer: LO6: D [2]
7. Answer: LO6: B [2]
8. Answer: LO6: A [2]
9. Answer: LO7: B [2]
10. Answer: LO7: A [2]
11. Answer, LO8: D [2]
12. Answer, LO8: B [2]
13. Answer, LO8: C [2]
14. Answer, LO9: D [2]
15. Answer, LO10: A [2]

QUESTION 2: TRUE/FALSE [10]

1. Answer, LO1: False: B [2]
2. Answer, LO2: True: A [2]
3. Answer, LO3: True: A [2]
4. Answer, LO4: False: B [2]
5. Answer, LO5: False : B [2]
6. Answer, LO6: False: B [2]
7. Answer, LO7: True: A [2]
8. Answer, LO8: False: B [2]
9. Answer, LO9: True: A [2]
10. Answer, LO10: True: A [2]

QUESTION 3: CASE STUDY [22]

1. Answer, LO1 (Answers may vary in expression): The aim of this paper is to identify an early plant disease that may affect the growth of their respective species [3]

2. **Answer, LO1/2 (Answer may vary in expression):** The research gaps seek to enhance greater transparency for detecting diseases in plants, even before their symptoms appear clearly. [3]
3. **Answer, LO5:** (supporting answer may vary in expression): This study is quantitative. It uses ML and DL classification models (statistical) to improve the accuracy of plant disease detection. [4]
4. **Answer, LO1:** Accuracy [2]
5. **Answer, LO10:** No, they are not using real plants. ML and DL techniques to classify early detection [4]
6. **Answer, LO1/8/9:** This review provides a comprehensive explanation of DL models used to visualize various plant diseases. In addition, some research gaps are identified from which to obtain greater transparency for detecting diseases in plants, even before their symptoms appear clearly. [2]
7. **Answer, LO3:** All abstracts do not include references [2]

QUESTION 4: [28]

1. **Answer, LO2:** [4]
Primary sources of information are those that proclaim new findings or knowledge. When researchers produce new knowledge this is shared with the research community. These are used to advance the field of enquiry (discourse) via scholarly sources. That is, the publication is peer-reviewed to validate the findings and/or results. Primary sources can also include dissertations/thesis.

Secondary sources of information provide a simplification, summary or analysis of several primary sources. A *literature review* is a common secondary source of information. A literature review is a synthesis of literature to provide a view of the state of the field.

2. **Answer, LO3:** [2]
 An *in-text* reference is a citation to an existing work or a piece of existing work (e.g. chapter in a book). It must include as a minimum the author/s surname, year of publication and page number (if applicable).
3. **Answer, LO5/8:** [4]

Quantitative research usually aims to test or confirm theories or assumptions. The numerical data is transformed into statistics that quantify the defined variables under investigation. This results in generalizable facts stemming from a sample to a much larger population.

Qualitative research aims to conduct deep investigations and gather in-depth insights into topics that are not well understood. Qualitative research is generally exploratory. It attempts to understand the underlying reasons, opinions and motivations.

4. **Answer, LO6/9:** SoGoSurvey, SurveyMonkey, Google forms, Survey planet [4]
5. **Answer, LO8:** (Any two, 2 marks each): [4]
- **Simple random sampling:** Every element has an equal opportunity of being selected. [2]
 - **Stratified sampling.** The population is split into groups. Elements are then randomly selected from each group. [2]
 - **Cluster sampling.** The population is split into groups. The researcher selects groups at random. Every element from the selected groups form the sample. [2]
 - **Systematic random sampling:** The population is ordered into a list. A starting point is selected at random and then every n th element is chosen for the sample.
6. **Answer, LO9:** [2]
They do not have a ranking and one response cannot be better than the other.
7. **Answer, LO10:** [8]
- i. Sensitive data
 - ii. Personal data
 - iii. Not personal data
 - iv. Sensitive data